

Video Game Film Adaptations

Box office performance, critical reception, and franchise fatigue — analyzed across nearly four decades of adaptations (1986–2024).

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PennState

Three Core Research Questions

Critic–Audience
Agreement Over
Time

Do critics and
audiences agree
on quality, and
has the gap
changed over
time?

Publisher Size &
Box Office

Does a game
publisher's annual
revenue predict
its film
adaptation's box
office success?

Franchise Fatigue

Do later franchise
installments earn
less and receive
worse critical
reception?

Data & Methodology

Primary Dataset

- TidyTuesday *Game Films* dataset, 439 films scraped from Wikipedia, spanning 1986–2024 with box office, budget, Rotten Tomatoes, Metacritic, and CinemaScore data.

246

Total Films

89

With Box Office Data

Cleaning Step

- Currency standardized to USD using fixed exchange rates
- All monetary values adjusted to constant 2024 dollars via CPI-U
- Filtered to theatrical releases only
- CinemaScore letter grades converted to 0–100 numeric scale

73

With RT Score

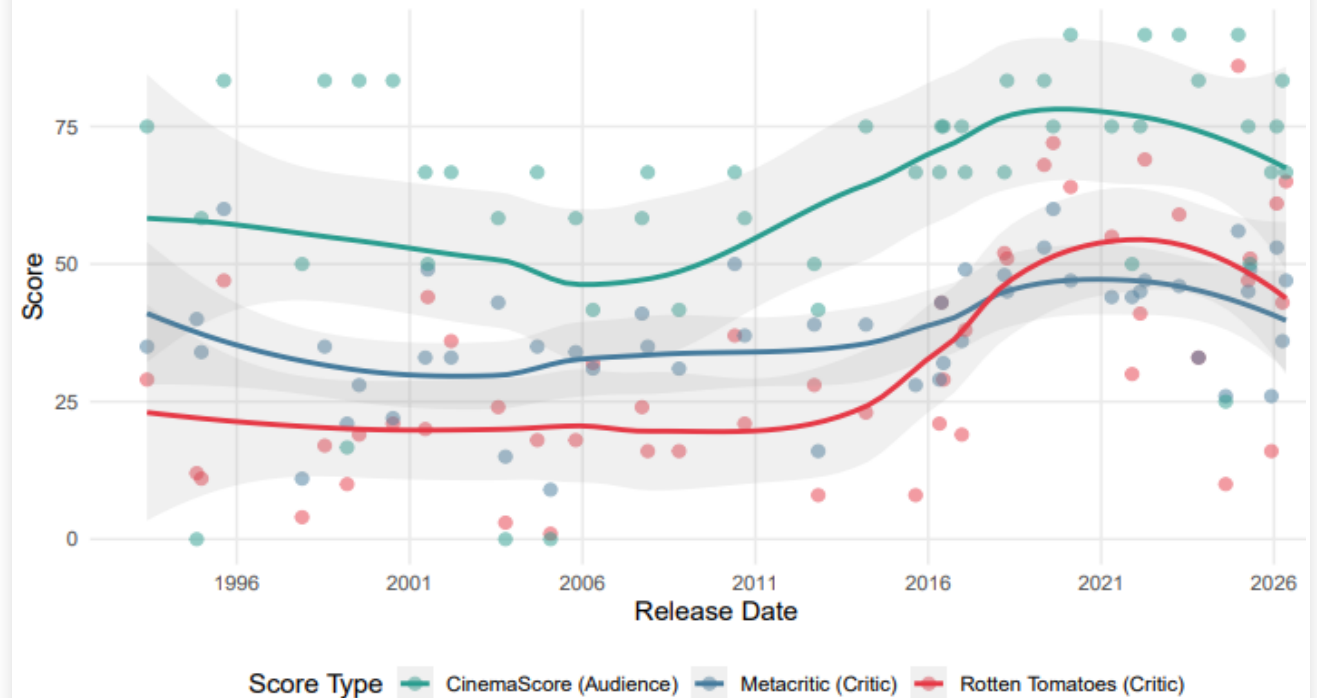
51

With CinemaScore

Critics vs. Audiences: A Persistent Gap

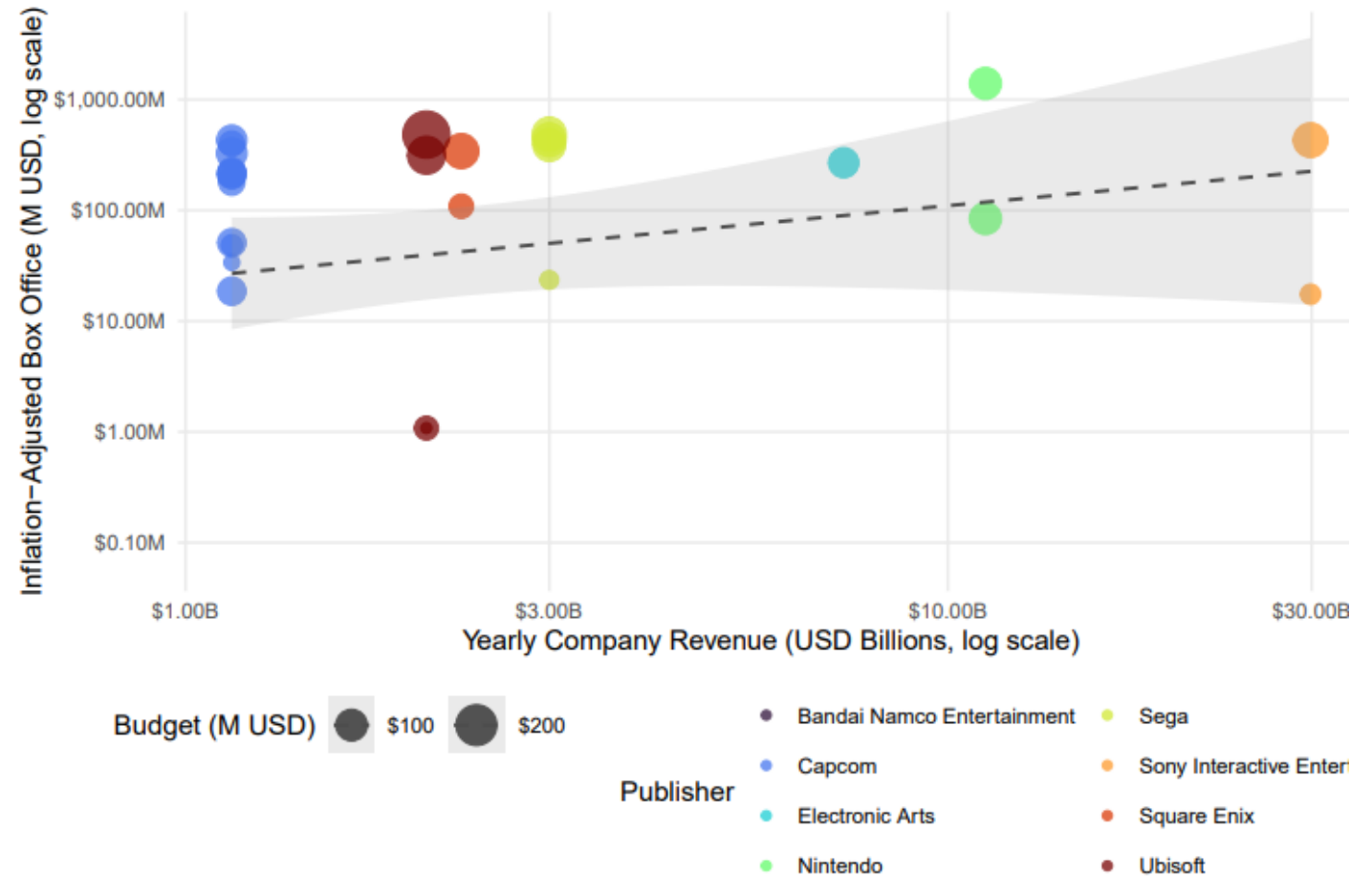
What the Data Shows

- Audiences consistently rate video game films **10–20 points higher** than critics across the entire timeline. CinemaScore trends hover at 60–70, while Rotten Tomatoes and Metacritic cluster at 40–55.
- All three score types show a modest upward trend since ~2010, and the critic–audience gap has **narrowed modestly**, suggesting improving quality and alignment in the genre.



Does Publisher Size Predict Box Office Success?

Publisher revenue is a **weak predictor** of box office performance. While Nintendo's *Super Mario* films are outliers at the top, smaller publishers like Capcom and Square Enix have achieved comparable results.



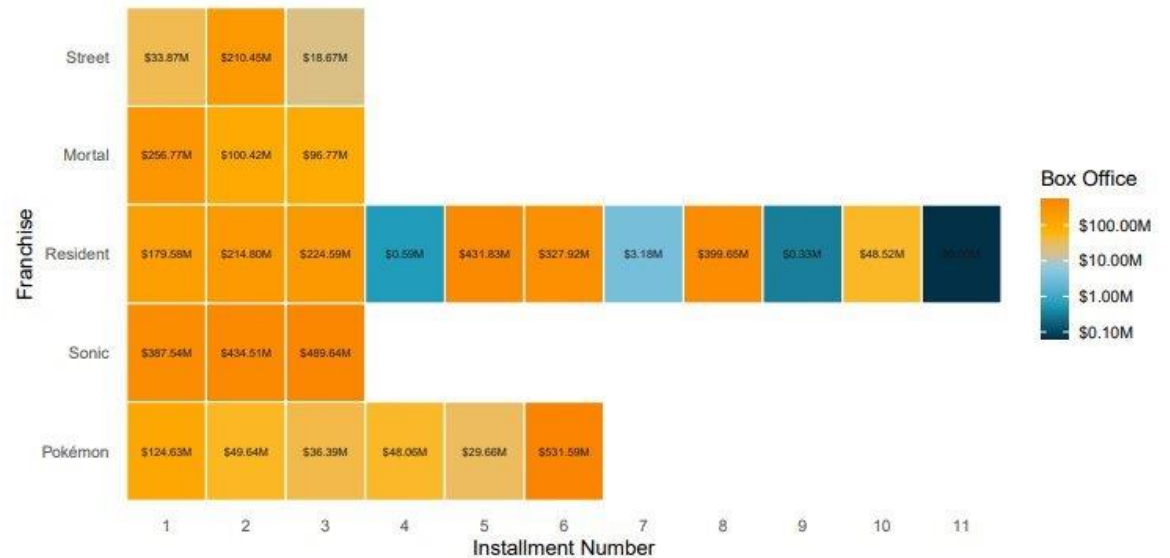
Franchise Fatigue Is Real

Box Office Decline

The overall trend across franchises shows a gentle downward slope in inflation-adjusted box office. *Resident Evil*, the most prolific franchise with six theatrical films shows a clear downward trajectory from ~\$150M to ~\$10M.

Critical Reception Falls Faster

Rotten Tomatoes scores decline **more steeply** than box office across installments. Even when later entries maintain revenue through fan loyalty, critics grow harsher. Studios should anticipate diminishing returns with each sequel.

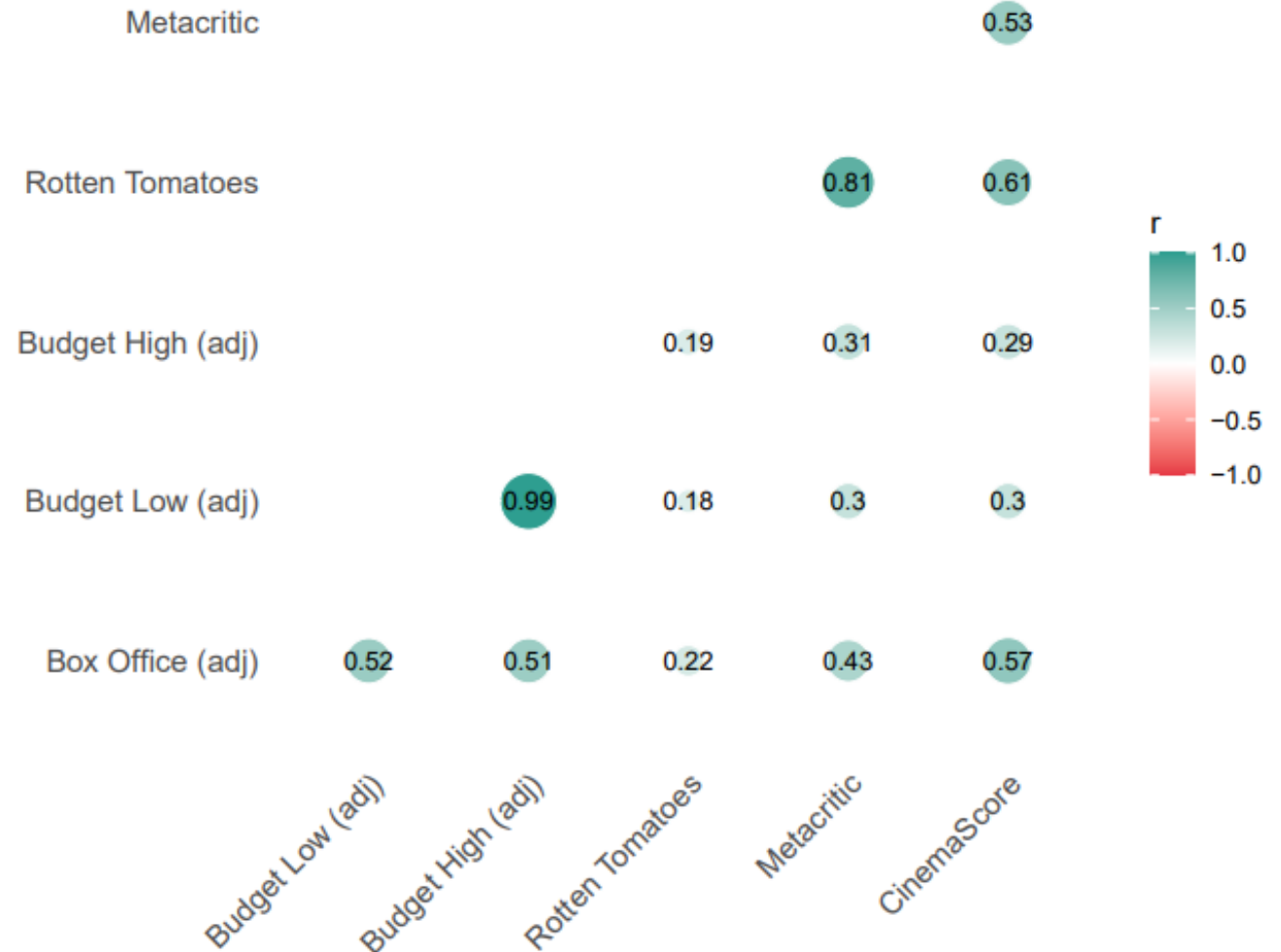


Correlation Insights

What Correlations Reveal

- **Budget and Box Office:** Moderate positive correlation ($r \approx 0.4$ – 0.5) — bigger budgets tend to earn more, but far from guaranteed
- **RT and Metacritic:** Strong correlation ($r \approx 0.85$) — both aggregate critic reviews
- **Budget and CinemaScore:** Weakest link ($r \approx 0.11$) — spending more does not meaningfully improve audience ratings

Correlation Matrix: Box Office, Budget & Scores



Limitations & Future Directions

Key Limitations

- Fixed exchange rates, historical daily rates would improve non-USD conversions
- Annual CPI-U averages, monthly data would refine inflation adjustment
- Missing data biases toward larger, well-documented releases
- First-word franchise heuristic is imperfect for multi-word titles
- All analyses are correlational, no causal claims

Future Work

- Incorporate genre and rating data to control for film characteristics
- Build predictive box office models using publisher, budget, and critic variables
- Analyze impact of global release strategies on performance
- Examine whether game-to-film release timing affects outcomes